

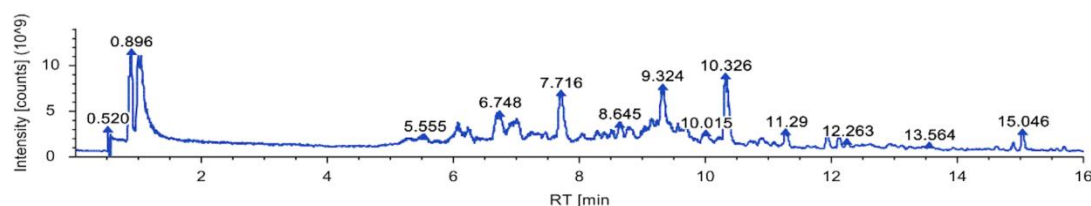
The report includes explanations of relevant terms.

Liquid chromatography-mass spectrometry

Liquid chromatography-mass spectrometry (LC-MS) is a technique that combines the separation capabilities of chromatography with the qualitative capabilities of mass spectrometry to detect different substances. Chromatography serves as the separation system, while mass spectrometry serves as the detection system.

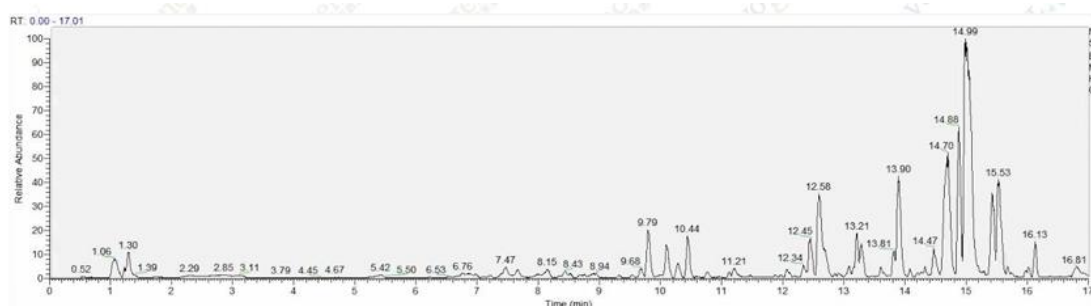
TIC

Total ion chromatogram (TIC) refers to a graph of the sum of all ion intensities over time within a selected mass range, which is the sum of all ion intensities in the mass spectrum at each time point.



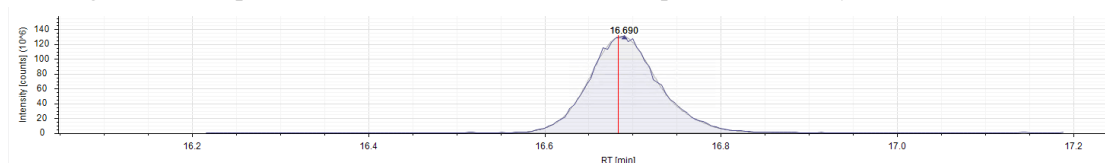
BPC

A base peak chromatogram (BPC) is a spectrum obtained by continuously plotting the intensity of the strongest ion at each time point in a mass spectrum.



XIC

An extracted ion chromatogram (XIC) is a chromatogram of parent ions with a specific mass-to-charge ratio. The peak area of the XIC can be used for quantitative analysis.



Pos, Neg scan mode

These refer to the positive ion mode (pos) and negative ion mode (neg), respectively. Due to differences in the chemical properties of metabolites, some metabolites tend to carry a positive charge during mass spectrometry ionization, while others tend to carry a negative charge. Amphiphilic metabolites can carry either a positive or a negative charge. Therefore, mass spectrometry commonly uses two scanning modes—positive ion scanning and negative ion scanning—to collect data during the detection process.

CV

The coefficient of variance (CV) is the ratio of the standard deviation to the mean. It is a statistical measure used to assess the degree of variation among observations in a sample. It reflects the

dispersion of data and can be used to determine the quality of stability.

Resolution

Resolution refers to the ability of a mass spectrometer to distinguish between two ions with similar masses. Calculation formula: Resolution = mass-to-charge ratio of an observed mass spectrum peak/peak width at half peak height.